4.9 Modeling with Polynomial Functions

Name _____ Period ____ Date _____

Algebra II 4.9 · Student sheet

I can…

- write cubic functions given the x-intercepts and a point.
- use finite differences to find the degree of a polynomial model.
- use technology to find polynomial models for real-life data.

Vocabulary · fill in the missing word

Finite differences — differences of _____ y-values when the x-values are equally spaced

Degree from differences — if the nth differences are _____ (and nonzero), the data fit a polynomial of degree n

Regression — _____ that finds the polynomial of a given degree that best fits data

Part 1 · Stacking cubes

Square pyramids of cubes: 1 layer = 1 cube, 2 layers = 1 + 4 cubes, 3 layers = 1 + 4 + 9 cubes, …

Layers	1	2	3	4	5	6
Cubes						
First differences						
Second differences						
Third differences						

Degree of the model: _____

Part 2 · Notes and examples (fill in each note, then work the example)

Cubic from x-intercepts: Cubic from x-intercepts p, q, r: f(x) = _____. Use another point to find _____.

Example 1

Write the cubic function whose graph has x-intercepts −2, 1, and 3 and passes through (0, −6).

Finite differences: Constant nth differences → degree _____.

Example 2

Use finite differences to find the degree of the polynomial function that fits the data. Then find the function.

x	1	2	3	4	5	6
f(x)	2	9	28	65	126	217

Example 3

Write the cubic function with x-intercepts -1, 2, and 4 that passes through (1, 12).

Part 3 · Partner practice

Partner 1

Find the degree of the polynomial that fits the data: $x = 1, 2, 3, 4, 5$; $y = 3, 6, 11, 18, 27$.

Partner 2

Write the cubic function with x-intercepts -3, 0, and 2 that passes through $(1, -8)$.